

ELECTROCARDIOGRAM (ECG)

Headings

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- **Electrocardiography** : Technique by which electrical activities of the heart are studied.
- **Electrocardiograph** : Instrument (machine) by which electrical activities of the heart are recorded.
- **Electrocardiogram** : Record or graphical registration of electrical activities of the heart, which occur prior to the onset of mechanical activities.

USES OF ECG

- Useful in determining and diagnosing:
 - a. Heart rate
 - b. Heart rhythm
 - c. Abnormal electrical conduction
 - d. Poor blood flow to heart muscle (Ischemia)
 - e. Heart attack (Myocardial infarction)
 - f. Coronary artery disease
 - g. Hypertrophy of heart chambers

ECG LEADS

- A total of **12 leads** are used for recording the electrical activity of the heart.
- ECG is recorded by placing a series of electrodes on the surface of the body (*called ECG leads*).
- **Einthoven's Triangle** : The heart is considered to be at the center of an imaginary equilateral triangle formed by connecting the roots of the **Right Arm (RA)**, **Left Arm (LA)**, and **Left Leg (LL)**.

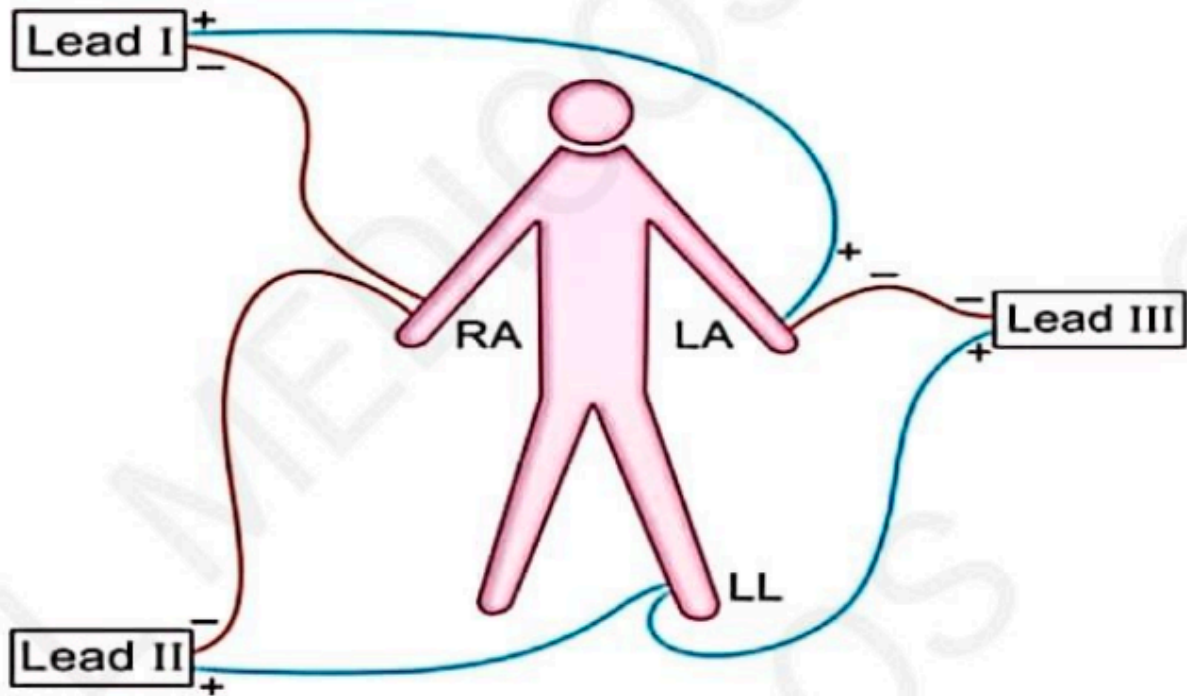


FIGURE 94.1: Position of electrodes for standard limb leads
RA = Right arm, LA = Left arm, LL=Left leg.

CLASSIFICATION OF ECG LEADS

ECG leads are broadly classified into **2 categories**:

- a. Bipolar Leads (Standard Limb Leads)
- b. Unipolar Leads (Augmented Limb Leads & Chest Leads)

1. Bipolar Limb Leads (Standard Limb Leads)

- Both electrodes are **active recording electrodes**.

Standard Lead	Negative Terminal (–)	Positive Terminal (+)
Lead I	Right Arm (RA)	Left Arm (LA)
Lead II	Right Arm (RA)	Left Leg (LL)
Lead III	Left Arm (LA)	Left Leg (LL)

2. Unipolar Leads

- Consist of one **active electrode (+ve)** and one **indifferent electrode** (*composite negative electrode*).

A. Unipolar Limb Leads (Augmented Voltage Leads):

- a. **aVR** : Active electrode on **Right Arm**
- b. **aVL** : Active electrode on **Left Arm**
- c. **aVF** : Active electrode on **Left Leg (Foot)**

B. Unipolar Chest Leads (Precordial 'V' Leads):

- a. V_1 : Right 4th intercostal space at right sternal margin
- b. V_2 : Left 4th intercostal space at left sternal margin
- c. V_3 : Midpoint between V_2 and V_4
- d. V_4 : Left 5th intercostal space on midclavicular line
- e. V_5 : Left 5th intercostal space on anterior axillary line
- f. V_6 : Left 5th intercostal space on midaxillary line

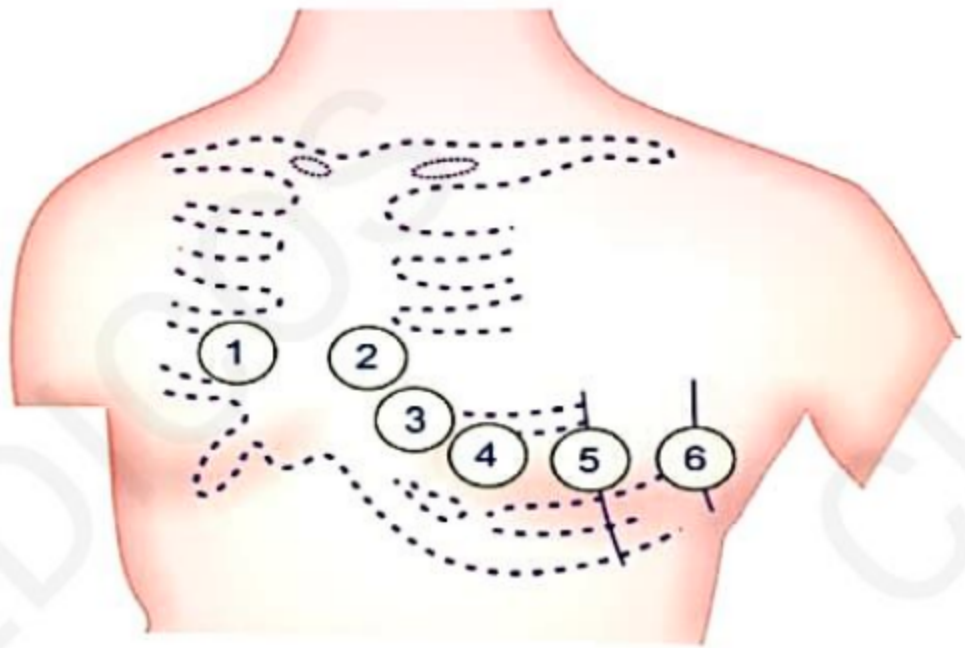


FIGURE 94.2: Position of electrodes for chest leads (V_1 to V_6)

WAVES AND COMPLEXES OF NORMAL ECG

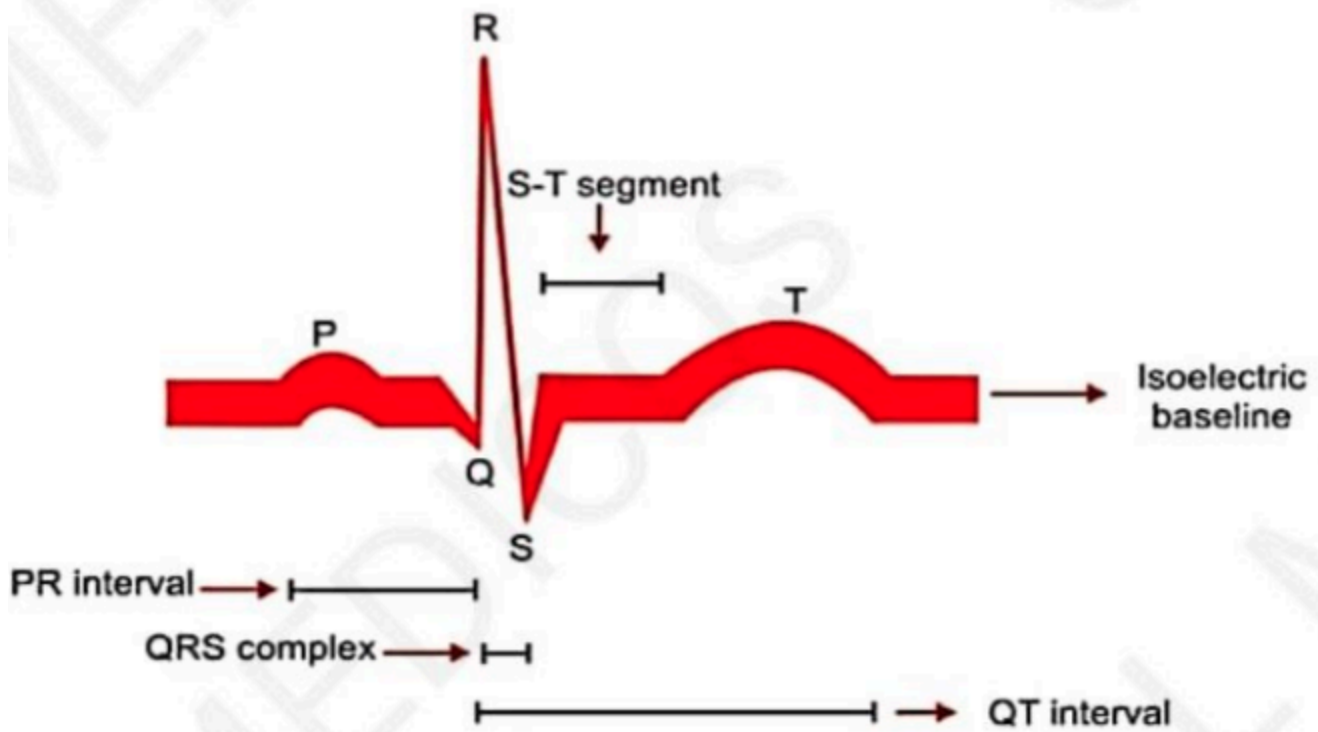


FIGURE 94.3: Waves of normal ECG

TABLE 94.1: Waves of normal ECG

Wave/Segment	From – To	Cause	Duration (second)	Amplitude (mV)
P wave	–	Atrial depolarization	0.1	0.1 to 0.12
QRS complex	Onset of Q wave to the end of S wave	Ventricular depolarization and atrial repolarization	0.08 to 0.10	Q = 0.1 to 0.2 R = 1 S = 0.4
T wave	–	Ventricular repolarization	0.2	0.3
P-R interval	Onset of P wave to onset of Q wave	Atrial depolarization and conduction through AV node	0.18 (0.12 to 0.2)	–
Q-T interval	Onset of Q wave and end of T wave	Ventricular depolarization and ventricular repolarization	0.4 to 0.42	–
S-T segment	End of S wave and onset of T wave	Isoelectric	0.08	–

(Waves recorded in Limb Lead II are considered typical)

Wave / Complex	Physiological Event	Clinical Significance / Pathological Changes
P-Wave (Atrial Complex)	Produced by Atrial Depolarization	• Tall P-wave: Right Atrial Hypertrophy • Small / Absent P-wave: Hyperkalemia • Inverted P-wave: Sinoatrial (SA) Block
QRS Complex (Ventricular Depolarization)	• Q-wave: Depolarization of basal portion of interventricular septum (IVS) • R-wave: Depolarization of apical portion of IVS & ventricles • S-wave: Depolarization of basal portion of ventricular muscle	• Prolonged QRS duration: – Hyperkalemia – Bundle branch block (BBB)
T-Wave (Ventricular Repolarization)	Represents Ventricular Repolarization	• Tall & Broad T-wave: Acute Myocardial Ischemia • Tall & Tented (Peaked) T-wave: Hyperkalemia • Small, Flat, or Inverted T-wave: Hypokalemia

Wave / Complex	Physiological Event	Clinical Significance / Pathological Changes
U-Wave (<i>Inconstant</i>)	Produced due to repolarization of papillary muscles (<i>not always seen</i>)	• Prominent U-wave: Hypercalcemia, Hypokalemia • Inverted U-wave: Myocardial Ischemia

Note : There is **no separate wave representing atrial repolarization** because it coincides with and is obscured by the much larger **QRS complex**.

INTERVALS AND SEGMENTS

Interval / Segment	Definition / Normal Features	Clinical Significance
1. P-R Interval	Time taken from the start of atrial depolarization to the start of ventricular depolarization	• Prolonged P-R Interval: Bradycardia, 1st Degree Heart Block • Shortened P-R Interval: Tachycardia (or Wolff-Parkinson-White syndrome)
2. Q-T Interval	Represents electrical depolarization and repolarization of ventricles (ventricular systole)	• Prolonged Q-T Interval: Myocardial Infarction, Myocarditis, Hypocalcemia • Shortened Q-T Interval: Hypercalcemia
3. S-T Segment	Isoelectric line from end of S-wave to beginning of T-wave (plateau phase of ventricular AP)	• ST Elevation: Acute Pericarditis, Acute Anterior / Inferior Myocardial Infarction (STEMI) • ST Depression: Acute Myocardial Ischemia, Posterior MI, Hypokalemia
4. R-R Interval	• Time interval between two consecutive R-waves • Normal duration = 0.8 seconds • Signifies the duration of one complete cardiac cycle	• Used to calculate Heart Rate: $HR = \frac{60}{\text{R-R interval in seconds}}$ • Evaluates Heart Rate Variability (HRV) and rhythm regularity